

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

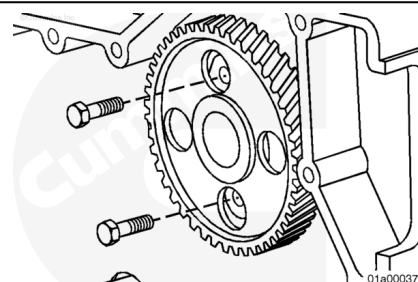
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the air intake piping from the left and right bank aftercoolers.
- Remove the rocker lever assemblies. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/57/57-003-009-tr.html>).
- Remove the push rods. Refer to Procedure 004-014 in Section 4 (</qs3/pubsys2/xml/en/procedures/57/57-004-014-tr.html>).
- Remove the damper and pulley. Refer to Procedure 001-052 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-052-tr.html>).
- Remove the gear cover and all related components. Refer to Procedure 001-031 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-031-tr.html>).
- Remove all of the cam follower assemblies. Refer to Procedure 004-001 in Section 4 (</qs3/pubsys2/xml/en/procedures/57/57-004-001-tr.html>).
- Remove the camshaft idler gear. Refer to Procedure 001-036 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-036-tr.html>).
- Remove the camshaft cover plate. Refer to Procedure 001-011 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-011-tr.html>).

Remove

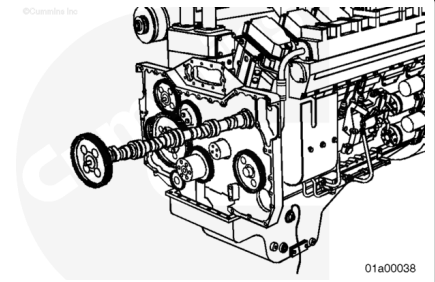
Rotate the camshaft to align the holes in the camshaft gear with the capscrews.

Remove the two capscrews.



⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



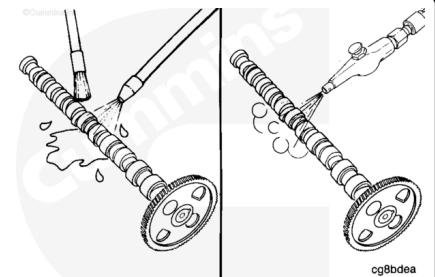
Use one hand to rotate and pull the camshaft slowly from the cylinder block and the other hand to balance the camshaft as it is removed.

Note : Support the camshaft through the camshaft follower cavities during the removal to prevent damage to the camshaft.

Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear protective clothing to reduce the possibility of personal injury.



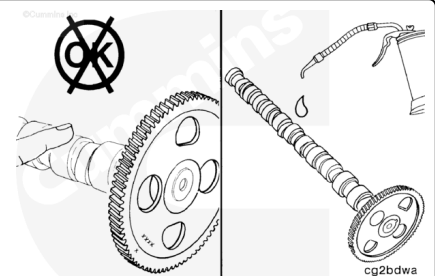
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the camshaft with solvent. Dry with compressed air.

⚠ CAUTION ⚠

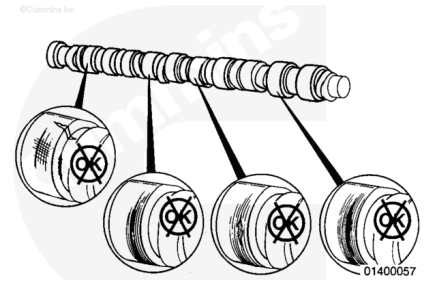
After the camshaft has been cleaned, do not touch the machined surfaces with bare hands. This can cause rust to form, which can damage the camshaft. Lubricate the camshaft with clean 15W-40 oil before handling.



Inspect for Reuse

⚠ CAUTION ⚠

Do not repair the camshaft by grinding the lobes. Damage to the component can occur.



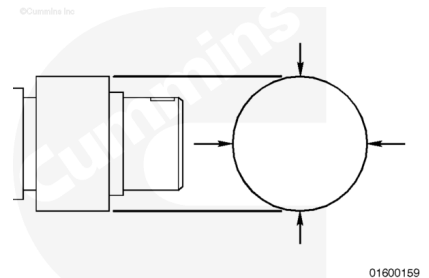
Inspect the valve lobes and bearing journals for cracking, pitting, or scoring. If the lobes or journals are damaged, the camshaft **must** be replaced. Refer to Camshaft Reuse Guidelines, Bulletin 3666052 (</qs3/pubsys2/xml/en/bulletin/3666052.html>).

Cummins Inc. does not recommend repairing camshafts by grinding the valve or injector lobes.

Note : Marks on the lobes or journals that are not felt with a fingernail are acceptable.

Camshaft Bushing Journal Outside Diameter

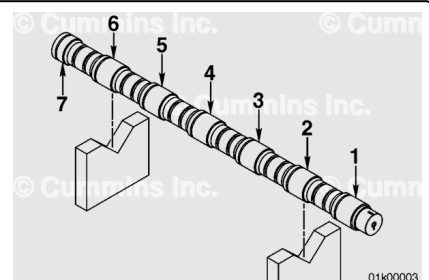
mm		in
64.948	MIN	2.557
64.984	MAX	2.558



If the diameter is **not** within specifications, the camshaft **must** be replaced.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

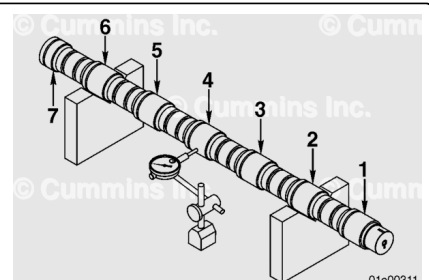


Place two vee blocks on a flat surface.

Support the camshaft on the vee blocks at journals 2 and 6.

Position a dial indicator so the stem touches the center line of the journal. Rotate the camshaft and measure the total indicator runout for each journal.

The dial indicator must be positioned at the centerline of any journal that is measured.



Torque Value: 0.03 mm [0.001 in]

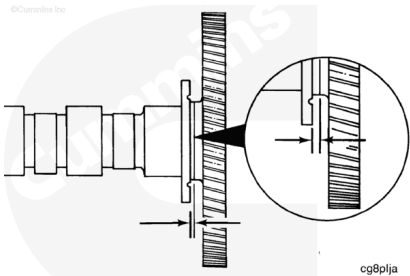
If the camshaft is **not** within specifications, it **must** be replaced.

Note : The max journal run-out applies to each journal.

Note : The two supported journals do not need to be measured.

Measure the clearance between the gear hub and the thrust plate.

Camshaft Thrust Plate Clearance		
mm		in
0.102	MIN	0.004
0.254	MAX	0.010

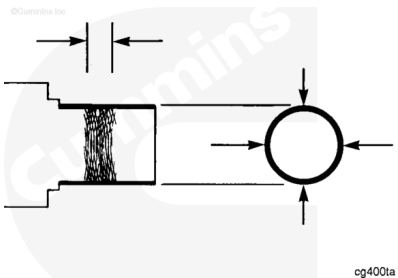


If the end clearance is **not** within specifications, the gear **must** be removed and inspected. Refer to Procedure 001-013 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-013-tr.html)

Clean and check the camshaft and camshaft gear for reuse. Check for fretting damage. The camshaft **must** be replaced if the fretting damage is more than 3 mm [1/8 in] wide.

Measure the outside of the camshaft.

Camshaft Outside Diameter (gear location)		
mm		in
44.068	MIN	1.735
44.084	MAX	1.736

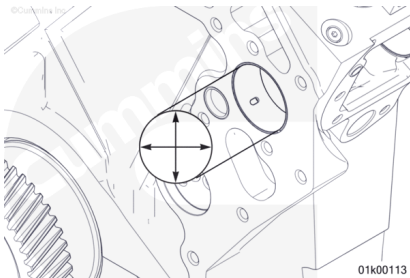


If the camshaft outside diameter is **not** within specifications, it **must** be replaced.

Inspect the amount of copper exposure on the camshaft bushing inside diameter. If it extends more than 180 degrees around the bushing, the bushing **must** be replaced.

Measure the inside diameter. Refer to Procedure 001-010 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-010-tr.html)

If the bushing is **not** within specifications, the bushing **must** be replaced.



Magnetic Crack Inspect

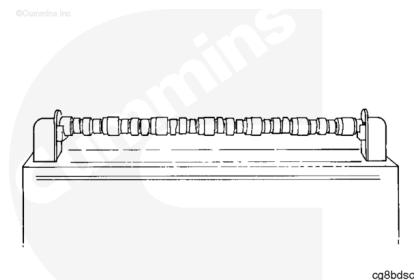
The gear and the thrust plate **must** be removed before performing this task. Refer to Procedure 001-013 in Section 1. (</qs3/pubsys2/xml/en/procedures/57/57-001-013-tr.html>)

Note : Marks on the lobes or journals that are not felt with a fingernail are acceptable.

Use a magnetic particle testing machine.

Use the head shot method:

- Adjust the machine to 2000 amps direct current or rectified VAC.
- Use the continuous method. Do **not** wet more than 1/3 of the camshaft at a time.
- Check the camshaft for cracks.

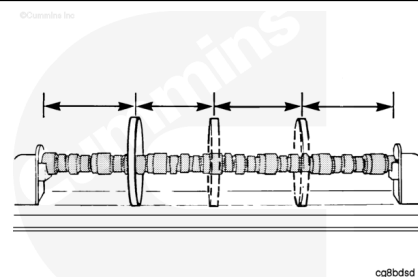


When the coil shot method is used, use a coil that is a minimum of 305 mm [12 in] diameter.

Ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

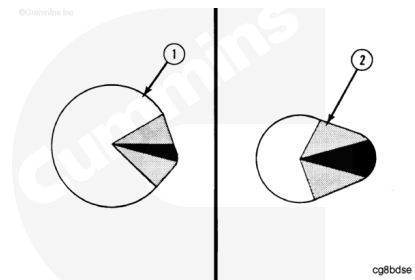
Use a coil with a minimum of 3600 amps direct current or rectified VAC and a maximum of 4000 amps direct current or rectified VAC.

Check the camshaft for cracks.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



An open indication is visible after the wetting operation has been completed. An indication below the surface is **not** visible after the wetting operation has been completed. An indication below the surface is visible with the use of an ultraviolet light.

Do **not** use the camshaft if the bearing journals have:

Bearing Journal Limits of Acceptance

- More than four open indications in an axial direction on any one bearing journals
- More than one half of the open indications extend more than one half the distance across the bearing journals
- Open indications in a circumferential direction.

The camshaft can be used if there are indications below the surface of the bearing journals.

Do not use the camshaft if the lobes have:

Camshaft Lobe Limits of Acceptance - Open Indications

- Open indications in a circumferential direction
- Open indications in the black or shaded areas (2)
- Open indications that are longer than 6 mm [0.236 in]
- Open indications that are closer than 5 mm [0.197 in] from the edge
- More than two indications on one of the lobes.

Camshaft Lobe Limits of Acceptance - Indications Below the Surface

- More than two indications in the shaded area (2)
- Below the surface indications in shaded area (2) that are longer 16 mm [0.630 in]
- More than three indications in the white area (1)
- Below the surface indication in the black area (2) on the injector area
- More than two indications in the black area on the valve lobe (2)
- Below the surface indications in the black area (2) of the valve lobe that are longer than 3 mm [0.118 in].

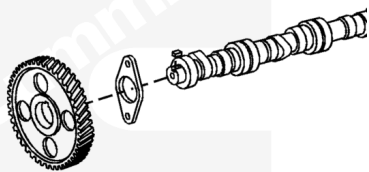
Demagnetize the camshaft.

Note : Camshaft must be demagnetized or engine damage may occur.

Clean the camshaft with solvent.

Install

Install the thrust plate and the camshaft gear on the camshaft, if necessary. Refer to Procedure 001-012 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-012-tr.html>).

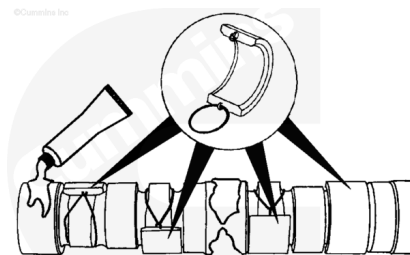


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Inspect the cam bushings for damage or cleanliness.

Check that all oil holes are aligned.

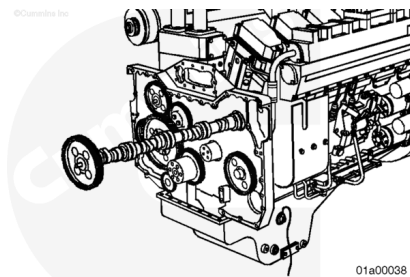
Use Lubriplate™ 105, or equivalent, to lubricate the camshaft and camshaft bushings.



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⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



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⚠ CAUTION ⚠

The right and left bank camshafts have different part numbers.

⚠ CAUTION ⚠

If the installation pilot is not available, use caution to reduce the possibility of the camshaft lobes damaging the bushings.

⚠ CAUTION ⚠

Support the camshaft through the camshaft follower cavities during installation to reduce the possibility of damage to the camshaft.

Install the camshaft.

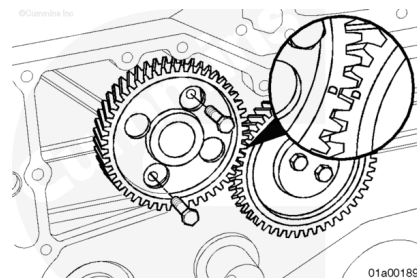
Rotate the camshaft **clockwise** and **counterclockwise** as it is being pushed.

Right Bank

Align the D mark on the camshaft gear with the D mark on the (small) idler gear.

Align the thrust bearing plate, and install the two capscrews.

Torque Value: 88 n•m [65 ft-lb]



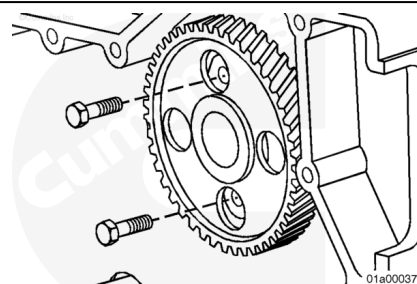
Left Bank

Align the thrust bearing plate.

Install the two capscrews.

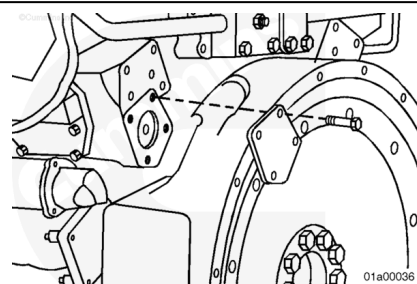
Note : The left bank camshaft marks will not be aligned until the left bank idler gear assembly is installed.

Torque Value: 88 n•m [65 ft-lb]



Install the camshaft rear cover and gasket. Refer to Procedure 001-011 in Section 1.

(/qs3/pubsys2/xml/en/procedures/57/57-001-011-tr.html)

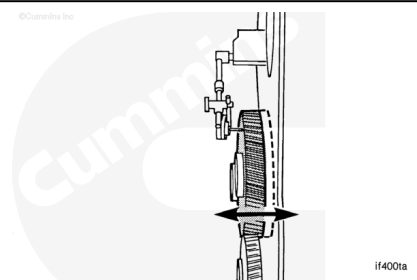


Use a dial indicator to measure the camshaft end clearance.

Refer to Procedure 001-065 in Section 1.

(/qs3/pubsys2/xml/en/procedures/57/57-001-065.html)

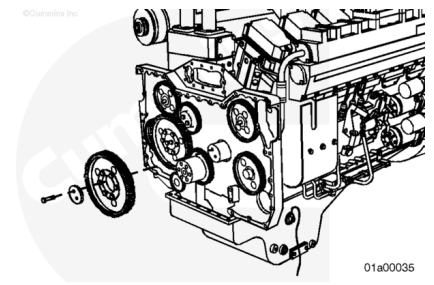
If the clearance is **not** within specifications, inspect for foreign material or a piece of gasket between the thrust bearing plate and the block, or heavy preservative on the surfaces.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the camshaft idler gear. Refer to Procedure 001-036 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-036-tr.html>).
- **Note : Verify the index marks on the camshaft and crankshaft are in alignment with the timing marks on the idler gear.**
- Measure the gear train backlash. Refer to Procedure 001-055 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-055.html>).
- Install the cam follower assemblies. Refer to Procedure 004-001 in Section 4 (</qs3/pubsys2/xml/en/procedures/57/57-004-001.html>).
- Install the gear cover and related components. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/57/57-001-031.html>).
- Install the damper and pulley. Refer to Procedure 001-052 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-052.html>).
- Install the push rods. Refer to Procedure 004-014 in Section 4 (</qs3/pubsys2/xml/en/procedures/57/57-004-014-tr.html>).
- Install the rocker lever assemblies. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/57/57-003-009-tr.html>).
- Install the air intake piping from the left and right bank aftercoolers. Refer to Procedure 010-002 in Section 10 (</qs3/pubsys2/xml/en/procedures/57/57-010-002-tr.html>).
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to 70°C [158°F] coolant temperature. Check for leaks.

